University of Ottawa

Department of Mathematics and Statistics

MAT 2143: Algebraic Structures

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Final Exam

April 26, 2019

Surname _____ First Name _____

Student # _____

Instructions:

(a) You have 180 minutes to complete this exam.

(b) All work to be considered for grading should be written in the space provided. The reverse side of pages is for scrap work. If you find that you need extra space in order to answer a particular question, you should continue on the reverse side of the page and indicate this **clearly**. Otherwise, the work written on the reverse side of pages will not be considered for marks.

(c) Write your student number at the top of each page in the space provided.

(d) No notes, books, scrap paper, calculators or other electronic devices are allowed.

(e) You are strongly recommended to write in **pen**, not pencil.

(f) You may use the last page of the exam as scrap paper.

(g) Cellular phones, unauthorized electronic devices or course notes (unless an open-book exam) are not allowed during this exam. **Phones and devices must be turned off and put away in your bag**. Do not keep them in your possession, such as in your pockets. If caught with such a device or document, the following may occur: you will be asked to leave immediately the exam and academic fraud allegations will be filed which may result in you obtaining a 0 (zero) for the exam.

By signing below, you acknowledge that you have ensured that you are complying with the above statement.

Signature _____

Please do not write in the table below.

Question	1	2	3	4	5	6	7	8	9	10	Total
Maximum	5	6	3	4	4	3	5	4	10	6	50
Grade											

1. Recall that $U(n)$ denotes the group formed by invertible elements of $\mathbb{Z}_n$.

(a) **[2 points]** Determine the number of elements of $U(12)$. You should justify your answer.

Solution: The number of elements is the number of integers in $\{1, \ldots, 12\}$ that are co-prime to 12, which is equal to 4.

(b) **[3 points]** Find the inverse of 5 in $U(18)$. You should justify your answer.

Solution: First we solve the equation $5x+18y = 1$. To do sowe follow the *gcd* method:

	3	1	1	
18	5	3	2	1

$$1 = 3 - (2 * 1) = 3 - (5 - 3 * 1)) = 3 * 2 - 5 = (18 - 5 * 3) * 2 - 5 = 18 * 2 - 5 * 7.$$

Thus $x = 7$ and $y = 2$ is a solution. Now it follows that $5 * 7 = 1$ in Z_{18}. Hence the inverse of 5 is 7.

2. (a) **[2 points]** Write down the definition of a cyclic group.

Solution: See Definition 6.1 in the notes.

(b) **[2 points]** Let $G = S_5$ and let $\sigma \in G$ be the permutation defined by

$$\sigma = (1\ 2\ 3)(4\ 5).$$

Let H be the cyclic subgroup of G that is generated by σ. List all of the elements of H concretely.

Solution:

$$\{(1\,2\,3)(4\,5), (1\,3\,2), (4\,5), (1\,2\,3), (1\,3\,2)(4\,5), \varepsilon\}$$

(c) **[1 point]** Can the subgroup H of part (b) be generated by an element other than σ? You should justify your answer.

Solution: Yes, for example it is generated by $(1\,3\,2)(4\,5)$.

(d) **[1 point]** Is the group $\mathbb{Z}_2 \times \mathbb{Z}_3$ cyclic? You should justify your answer.

Solution: Yes it is cyclic because it is generated by the element $(1, 1) \in \mathbb{Z}_2 \times \mathbb{Z}_3$

3. [**3 points**] Write down the proof of the following statement:

Let G be a group and let H and K be subgroups of G. Then $H \cap K$ is also a subgroup of G.

Solution: See the proof of Proposition 5.8 in the lecture notes

4. [**4 points**] For each of the following statements, write '**T**' if the statement is true, and write '**F**' if the statement is false.

You will receive .5 points for each correct answer, 0 points for each blank answer, and -.25 points for each incorrect answer. Your total score for this question will not be negative.

_____ The group $\mathbb{Z}_{24}$ has exactly 8 distinct subgroups.

_____ The composition of any two cycles in S_n is again a cycle of S_n.

_____ If $n \geq 5$, then the group S_n does not have any proper normal subgroups (i.e., other than $\{\varepsilon\}$ and S_n).

_____ Every group with 37 elements is cyclic.

_____ A group G with 100 elements cannot have a subgroup with 17 elements.

_____ The dihedral group D_7 contains an element g such that $o(g) = 2$.

_____ Every group G with 90 elements has a normal subgroup with 45 elements.

_____ For every finite abelian group G with n elements, we can find a surjective group homomorphism $\psi : \mathbb{Z}^n \to G$.

Solution: The answers are: T, F, F, T, T, T, T, T

5. (a) **[2 points]** Write down the cycle decomposition of the permutation $\sigma \in S_{10}$ given by

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \\ 5 & 2 & 7 & 8 & 3 & 4 & 1 & 6 & 10 & 9 \end{pmatrix}.$$

Then determine the order of σ. You should justify your answer.

Solution: The cycle decomposition is

$$(1\ 5\ 3\ 7)(4\ 8\ 6)(9\ 10)$$

and the order of the element is $lcm(4, 3, 2) = 12$.

(b) **[1 point]** Let $\tau \in S_9$ be the permutation given by the following product of cycles:

$$\tau = (1\ \ 7\ \ 5\ \ 2\ \ 4\ \ 3)(2\ \ 9\ \ 7).$$

Is $\tau \in A_9$? You should justify your answer.

Solution: Every cycle of length d is a product of d transpositions. It follows that τ is a product of $5 + 2 = 7$ transpositions, hence $\tau \notin A_9$.

(c) **[1 point]** Let τ be as in part (b). Does there exist a permutation $\eta \in S_9$ such that $\tau = \eta^2$? You should justify your answer.

Solution: No. If such a permutation exists then τ will be an even permutation because the product of two even or two odd permutations is always even but τ is odd.

6. (a) **[2 points]** Write down the definition of two conjugate permutations $\sigma, \sigma' \in S_n$.

Solution: See lecture notes.

(b) **[1 point]** Consider the permutations $\sigma, \sigma' \in S_5$ given by

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 2 & 1 & 5 & 4 \end{pmatrix} \quad \text{and} \quad \sigma' = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 4 & 5 & 1 & 3 \end{pmatrix}$$

Explain why σ and σ' are <u>not</u> conjugate permutations.

Solution: Two conjugate permutations have equal cycle lengths, but the cycle lengths of σ are 2 and 2, and the cycle lengths of σ' are 3 and 2, so they cannot be conjugate.

7. (a) **[2 points]** Write down the definition of a normal subgroup of a group G.

Solution: See Definition 11.1 in the lecture notes.

(b) **[3 points]** Write down the proof of the following statement:

Let G be a group and let H be a subgroup of G such that $[G : H] = 2$. Then H is a normal subgroup of G.

Solution: This is Proposition 11.6 in the lecture notes.

8. (a) **[1 point]** Write the definition of the kernel of a group homomorphism $\phi : G \to H$.

Solution: See Definition 12.4 in the lecture notes.

(b) **[2 points]** Write down the proof of the following statement:

Let $\phi : G \to H$ be a group homomorphism such that $\ker(\phi) = \{e\}$, where "e" denotes the identity element of G. Then ϕ is an injection.

Solution: This is Proposition 12.7 in the lecture notes.

(c) **[1 point]** Give an example of a group homomorphism $\phi : G \to H$ which is not an injection.

Solution: For example $G = \mathbb{Z}_2 \times \mathbb{Z}_2$, $H = \mathbb{Z}_2$, and $f : G \to H$ defined by $f(a, b) = a$ for all $(a, b) \in \mathbb{Z}_2 \times \mathbb{Z}_2$.

9. Let $G = \mathbb{Z}$ and let $H = \{m \in \mathbb{Z} \ : \ 9|m\} = \{m \in \mathbb{Z} \ : \ m \text{ is divisible by } 9\}$.

(a) [**4 points**] Prove that H is a normal subgroup of G. You need to prove two facts: H is a subgroup, and H is normal.

Solution: By the subgroup test, we need to verify that if $a, b \in H$ then $a - b \in H$. We have

$$a, b \in H \ \Rightarrow \ \exists a', b' \in \mathbb{Z} \ : \ a = 9a' \text{ and } b = 9b'$$

and therefore $a - b = 9(a' - b') \in H$.

To show H is normal, note that $\mathbb{Z}$ is abelian so every subgroup of $\mathbb{Z}$ is normal.

(b) [**2 points**] Describe the distinct left cosets of H in G concretely.

Solution: They are the sets $a + H$ for $a \in \{0, 1, 2, 3, 4, 5, 6, 7, 8\}$.

(c) [**2 points**] Find an integer d such that the quotient group G/H is isomorphic to $\mathbb{Z}_d$. Describe an explicit isomorphism $\phi : G/H \to \mathbb{Z}_d$.

Solution: We show that the map $f : G/H \to \mathbb{Z}_9$ defined by

$$f(a+H) := a \text{ for } a \in \{0,1,2,3,4,5,6,7,8\}$$

is an isomorphism (hence $d = 9$ is the answer). To verify this, note that the above map is a bijection, and a group homomorphism because

$$f((a+H)+(b+H)) = f(a+b+H) = a+b \text{ in } \mathbb{Z}_9,$$

and

$$f(a+H)+f(b+H) = a+b \text{ in } \mathbb{Z}_9.$$

(d) [**2 points**] Find a subgroup of the quotient group G/H with 3 elements. You should concretely identify the elements of this subgroup.

Solution: because of the isomorphims in part (c), such a subgroup of G/H should correspond to a subgroup of 3 elements in $\mathbb{Z}_9$. The latter group has a subgroup with 3 elements: $\{0,3,6\}$. The corresponding subgroup of G/H is $\{0+H, 3+H, 6+H\}$.

10. Let $\phi : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ be the map defined by

$$\phi(x, y) = -6x + 10y \text{ for all } x, y \in \mathbb{Z}.$$

(a) **[2 points]** Prove that ϕ is a group homomorphism.

Solution:

$$\begin{aligned}\phi((x, y) + (x', y')) &= \phi(x + x', y + y') = -6(x + x') + 10(y + y') \\ &= -6x + 10y - 6x' + 10y' = \phi(x, y) + \phi(x', y').\end{aligned}$$

(b) **[2 points]** Describe $\text{im}(\phi)$ concretely. You should justify your answer.

Solution: By Bezout's theorem, the image of ϕ is the set of integer multiples of $gcd(-6, 10) = 2$, hence the image of ϕ is $2\mathbb{Z}$.

(c) [**2 points**] Set $G = \mathbb{Z} \times \mathbb{Z}$ and $H = \ker(\phi)$. Use the First Isomorphism Theorem to justify that the quotient group G/H is isomorphic to $\mathbb{Z}$.

Hint: You do not need to compute $\ker(\phi)$ explicitly.

Solution: By the first isomorphism theorem the map ϕ is an isomorphism between $G/\ker\phi$ and the image of ϕ, which is $2\mathbb{Z}$. This $G/H \cong 2\mathbb{Z} \cong \mathbb{Z}$. For the last isomorphism we can take the isomorphism $phi : 2\mathbb{Z} \to \mathbb{Z}$ given by $2m \mapsto m$ for $m \in \mathbb{Z}$:

$$\phi(2m + 2m') = \phi(2(m + m')) = m + m' = \phi(2m) + \phi(2m').$$